

3blue1brown ring question solved

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Question:

Given any 10 arbitrary points in a 2D space, can we always place a collection of one or more disks of radius 1 (without any overlapping) such that every point lies inside at least one of the disks?

i.e.

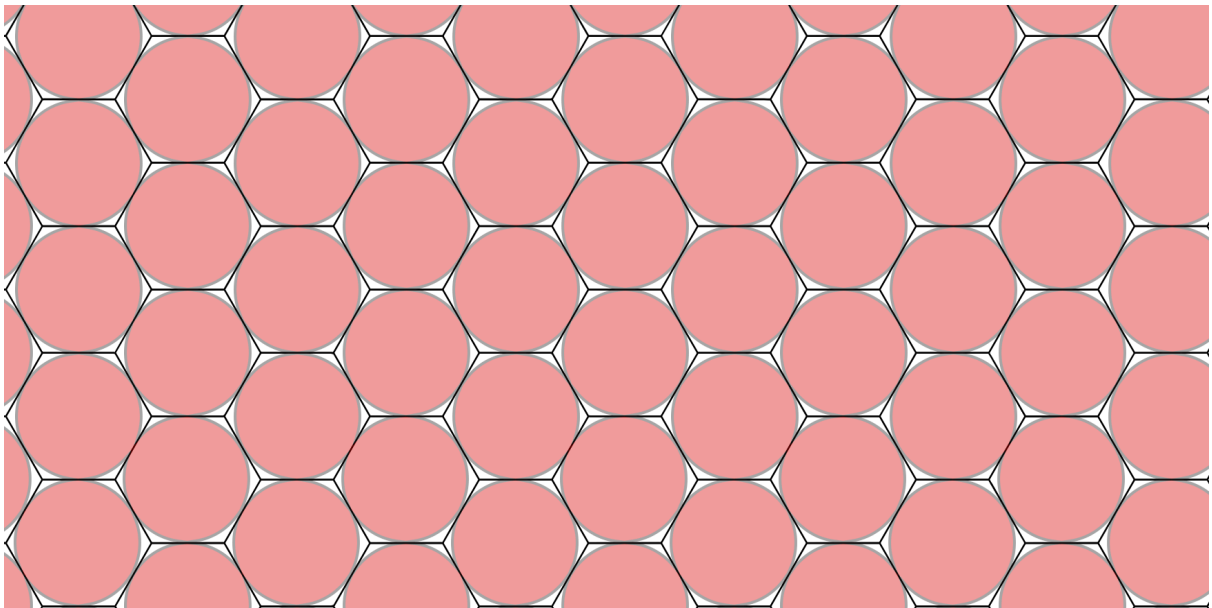
P is a set of 10 points and C is a set of center of circles with radius 1

$$\begin{aligned} \forall p \in P, \exists c \in C \mid \text{distance}(p, c) \leq 1 \\ \forall c_1, c_2 \in C \mid c_1 = c_2 \vee \text{distance}(c_1, c_2) > 2 \end{aligned}$$

Yes, we prove that it is **always possible** using a hexagonal tiling and an area argument.

1: Hexagonal grid of disks

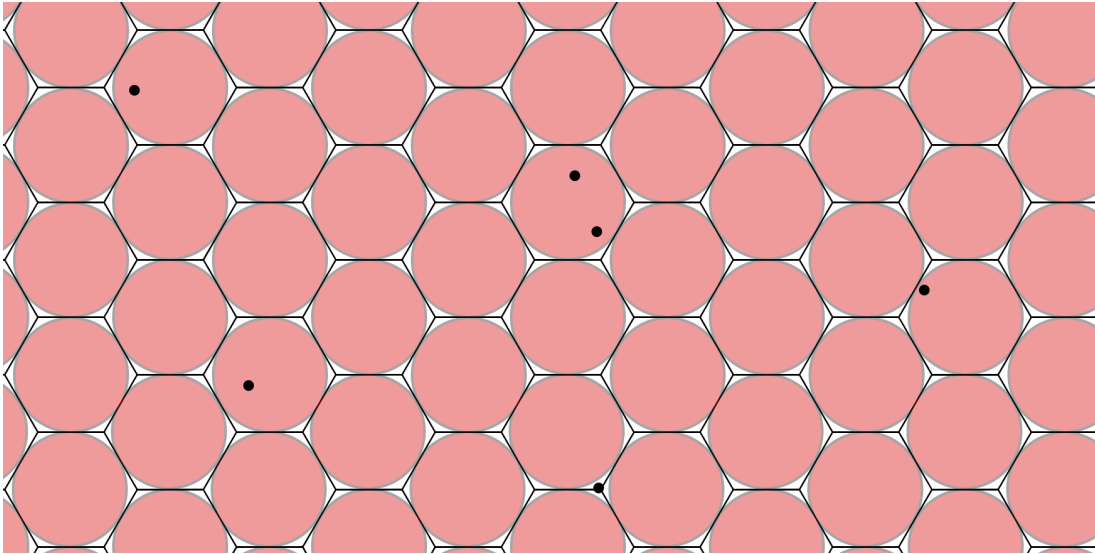
Let's consider an infinite tiling of the $\mathbb{R}^2$ space by regular hexagons. Inside each hexagon, there is inscribed a disk of radius $r = 1$. The disks do not overlap because the distance between the centers of two adjacent hexagons is exactly 2 (the disks are tangent).



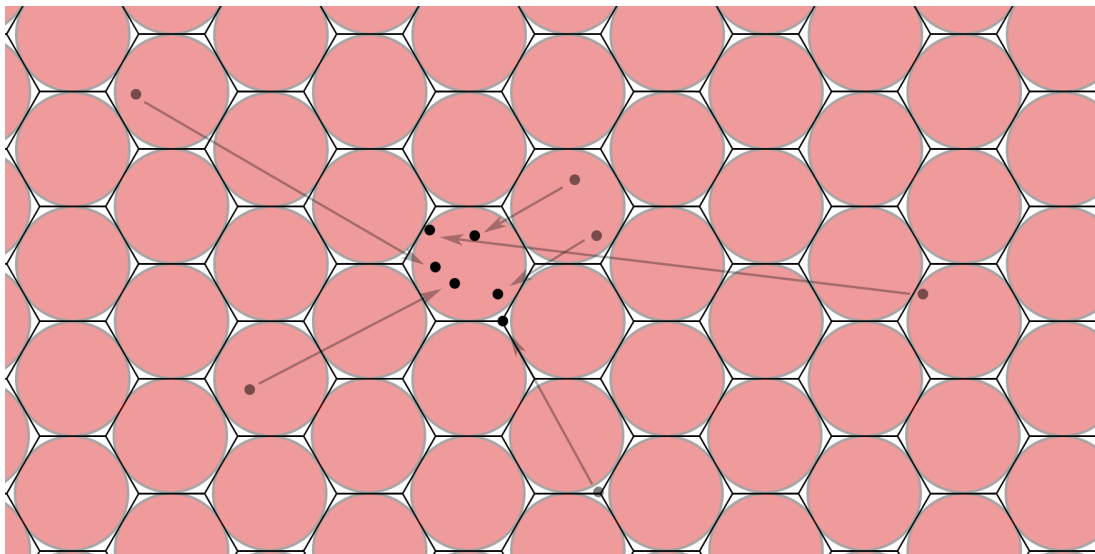
Now we can prove that for any 10 points, we can displace the grid of circles by a vector v such that all of the 10 points lie inside at least one circle.

2: Reducing to a single hexagon

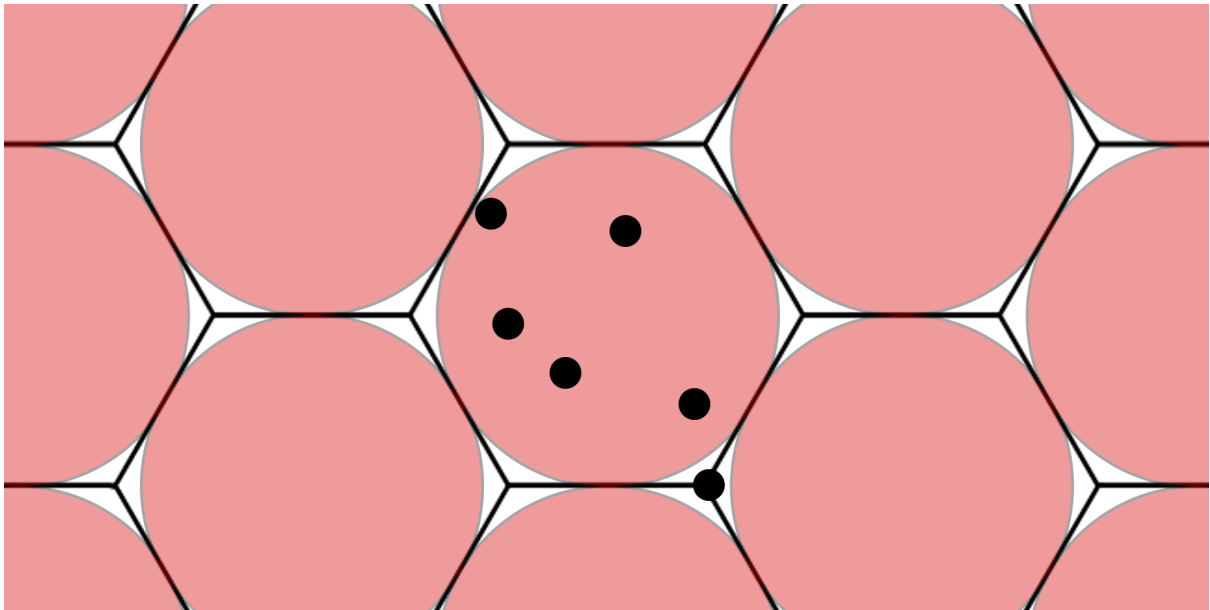
Let's take some arbitrary points (in this example we have six, but it works for any amount):



The arrangement is periodic. For any point in the plane, we look at its distance with the center of the hexagon that contains it. We can move any point from its hexagon to any other hexagon by just keeping the relative distance between the point and its associated hexagon's center the same (i.e. we add or subtract lattice vectors to the point). Thus, without loss of generality, we can assume all ten points (six in this example) lie inside the same reference hexagon.



Now we can work with a much neater set of points all reduced to a single hexagon.



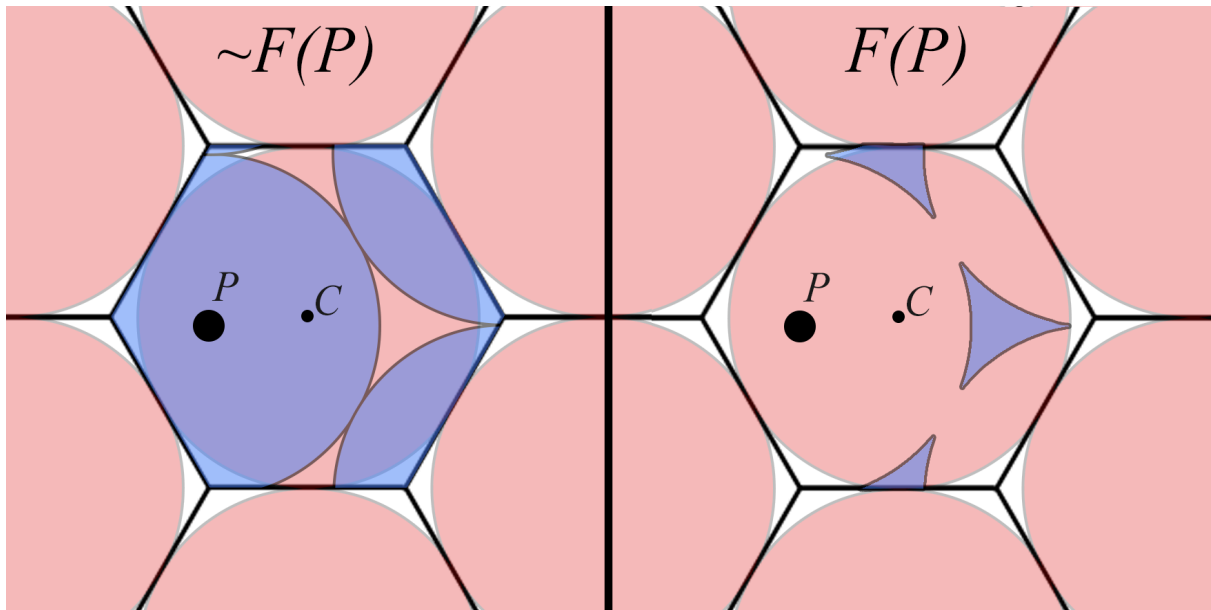
3: Shifting the grid as a variable

Now let the whole grid of circles be shifted by a vector v so the center of the reference circle (the one inscribed in the reference hexagon where all the points are) moves from the origin to v . The space of possible shifts v can be restricted to the reference hexagon itself, since shifting by a lattice vector gives an equivalent configuration.

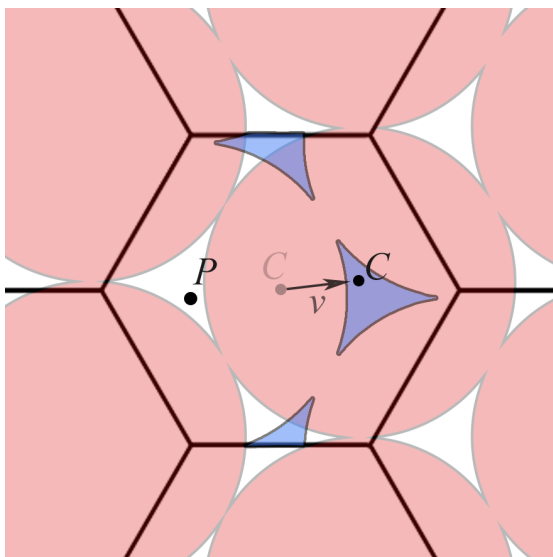
We now ask: *does there exist at least one possible shift v such that the 10 points are covered by some disk?*

4: Forbidden translations

Fix a point P inside the reference hexagon. Which shifts v cause P to be outside every disk? For P to be covered, there must exist a grid center (origin + v) at a distance ≤ 1 from P . The opposite condition (P is not covered) means that P is a distance > 1 from **all** grid centers. This defines a region $F(P)$ inside the reference hexagon, that we will call from now on the **forbidden region** for that point.



If we shift the grid by any v in $F(P)$, then point P will not be inside any disk. On the other hand, if we shift by a v in $\sim F(P)$, then P is covered.



As the image in the left shows, if the center of the grid of circles C is displaced by any vector v that lies on the forbidden region $F(P)$, its respective point P will now be not contained by any of the rings.

It can be shown (by elementary euclidean geometry) that the area of $F(P)$ is constant for any point P inside the hexagon.

Let:

- Area of reference hexagon: $A_h = 2\sqrt{3} \simeq 3.464$
- Area of forbidden region for one point: $A_f = A_h - \pi r^2 = A_h - \pi \simeq 0.323$

5: Union of forbidden regions

We have 10 points $P_1, \dots, P_{10}$, each with its forbidden region $F_1, \dots, F_{10}$ inside the reference hexagon. If the union $F_1 \cup \dots \cup F_{10}$ **does not** cover the whole hexagon, then there exists a shift v outside $\bigcup_{i=1}^{10} F_i$ meaning every point is covered. On the other hand, if the union **does** cover the whole hexagon, then no such shift exists - but we will show this cannot happen.

6: Area argument

The total area of the hexagon is $A_h \simeq 3.464$. Even in the worst possible case, where the forbidden regions are pairwise disjoint (which maximizes the covered area), the total area covered by the union is at most:

$$10 \times A_f = 10(2\sqrt{3} - \pi) \simeq 3.23$$

Since $10A_f < A_h$, the union cannot cover the entire hexagon. Therefore, no matter how the 10 points are placed, there will always be some shift v that avoids all forbidden regions.

7: Conclusion

For any 10 points, we can shift the infinite grid of circles (radius 1, non-overlapping) so that all points lie inside some disk. Hence, the answer to the problem is **yes**.